

```
<!DOCTYPE html><html lang="en"><head> <meta charset="UTF-8"><meta
name="viewport"content="width=device-width, initial-scale=1.0"><title>Calculus in Food Security: Oil Palm
Yield Optimization</title><!-- Tailwind CSS for modern, responsive UI styling --> <script
src="https://cdn.tailwindcss.com"></script><!-- Chart.js for data visualization --> <script
src="https://cdn.jsdelivr.net/npm/chart.js"></script></head><body class="bg-slate-50 text-slate-800 font-sans
antialiased min-h-screen flex flex-col"><!-- Header Navigation --><header class="bg-emerald-800 text-white
shadow-md"><div class="max-w-7xl mx-auto px-4 py-5 sm:px-6 lg:px-8 flex flex-col sm:flex-row justify-
between items-center gap-4"><div><h1 class="text-2xl font-bold tracking-tight">Bio-Calculus Interactive
Dashboard</h1><p class="text-emerald-200 text-sm mt-1">Application of Derivatives in Malaysian Food
Security & Precision Agriculture</p></div><div class="bg-emerald-700 px-4 py-2 rounded-full text-xs font-
semibold tracking-wide border border-emerald-600 uppercase">SFM: Food Security
Module</div></div></header><!-- Main Content Layout --><main class="flex-grow max-w-7xl w-full mx-auto
px-4 py-8 sm:px-6 lg:px-8 grid grid-cols-1 lg:grid-cols-3 gap-8"><!-- Left Column: Bioscience Scenario &
Formulas --><div class="lg:col-span-1 space-y-6"><!-- Scenario Card --><div class="bg-white p-6 rounded-2xl
shadow-sm border border-slate-200"><div class="flex items-center gap-2 text-emerald-700 mb-3"><svg
class="w-6 h-6 fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-
linejoin="round" stroke-width="2" d="M12 6.25v13m0-13C10.832 5.477 9.246 5 7.5 5S4.168 5 2.5 5.477 3
6.25v13C4.168 18.477 5.754 18 7.5 18s3.332 4.477 4.5 1.253m0-13C13.168 5.477 14.754 5 16.5 5c1.747 0
3.332 4.477 4.5 1.253v13C19.832 18.477 18.247 18 16.5 18c-1.746 0-3.332 4.477-4.5 1.253"></path></svg><h2
class="text-xl font-bold">The Bioscience Scenario</h2></div><h3 class="font-semibold text-slate-700 mb-
2">Optimizing Oil Palm Fresh Fruit Bunch (FFB) Yield Dynamics</h3><p class="text-slate-600 text-sm leading-
relaxed mb-4">In Malaysia, oil palm (Elaeis guineensis) is crucial to national agricultural GDP and
global food supply chains. A critical problem in plantation management is tracking the Fresh Fruit Bunch
(FFB) yield rate as a plantation ages.</p><p class="text-slate-600 text-sm leading-relaxed mb-4">Yields
typically follow a non-linear trajectory: rising sharply after planting, peaking during prime maturity, and slowly
declining as trees grow too tall for easy harvesting.</p><div class="p-3 bg-amber-50 border-l-4 border-amber-
500 rounded text-amber-900 text-xs"><strong>Why Derivatives Matter:</strong> Finding the exact year
where yield acceleration slows down (inflection points) and where yield reaches its absolute peak ( $f'(x) = 0$ )
allows plantation managers to plan replanting cycles and calculate precise fertilizer interventions.</div></div><!-- Mathematical Modeling Card --><div class="bg-white p-6 rounded-2xl shadow-sm border
border-slate-200"><div class="flex items-center gap-2 text-emerald-700 mb-4"><svg class="w-6 h-6"
fill="none" stroke="currentColor" viewBox="0 0 24 24"><path stroke-linecap="round" stroke-linejoin="round"
stroke-width="2" d="M9 7h6m0 10v-3m-3 3h.01M9 17h.01M9 14h.01M12 14h.01M15 11h.01M12 11h.01M9
11h.01M7 21h10a2 2 2 0 002-2V5a2 2 2 0 00-2-2H7a2 2 2 0 00-2 2v14a2 2 2 0 002 2z"></path></svg><h2
class="text-xl font-bold">Mathematical Modeling</h2></div><!-- Equations Display --><div class="space-y-4
text-sm"><div class="bg-slate-50 p-3 rounded-lg border border-slate-150"><span class="block text-xs font-
semibold text-slate-500 uppercase mb-1">Yield Function  $f(x)$ </span><div class="text-base font-mono text-
slate-800 text-center py-2 bg-white rounded border border-slate-200 shadow-inner"><math>f(x) = -0.015x^3 +
0.52x^2 - 1.8x + 3.5</math></div><p class="text-[11px] text-slate-500 mt-1.5">Models FFB Yield in Tonnes per
Hectare per Year (t/ha/yr) where  $x$  is the plantation age in years.</p></div><div class="bg-slate-50 p-3
rounded-lg border border-slate-150"><span class="block text-xs font-semibold text-slate-500 uppercase mb-
1">Derivative Function  $f'(x)$  (Rate of Change)</span><div class="text-base font-mono text-emerald-700 text-
center py-2 bg-white rounded border border-slate-200 shadow-inner"><math>f'(x) = -0.045x^2 + 1.04x - 1.8</math></div><p
class="text-[11px] text-slate-500 mt-1.5">Calculates the instantaneous growth velocity of yield in
(t/ha/yr) per year at any precise moment.</p></div></div><!-- Data Reference Card --><div
class="bg-white p-4 rounded-2xl shadow-sm border border-slate-200 text-xs text-slate-500"><span
class="font-semibold text-slate-700 block mb-1">Empirical Data Grounding:</span>This model is calibrated
against historical baseline benchmarks published by the Malaysian Palm Oil Board (MPOB) under their
periodic report webpage:<span class="block mt-1 font-mono text-emerald-800 bg-slate-150 p-1 rounded
```

```

select-all break-all">"MPOB Oil Palm Yield Performance and Age Profile Data"</span>and matching trends
from the **Department of Statistics Malaysia (DOSM)** agricultural productivity logs. </div></div><!-- Right
Column: Interactive Graph & Derivative Output Panel --><div class="lg:col-span-2 space-y-6"><!-- Graph
Visualization Window --><div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200 flex flex-col
h-full justify-between"><div><div class="flex justify-between items-start mb-2"><div><h2 class="text-xl font-
bold text-slate-800">Yield Trajectory Analysis</h2><p class="text-xs text-slate-500">Click anywhere inside the
chart plot area to analyze calculus properties at that specific tree age.</p></div><span class="text-xs bg-slate-
100 text-slate-600 px-2.5 py-1 rounded-md border border-slate-200">Interactive Plot</span></div><!-- Chart
canvas wrapper --><div class="relative w-full min-h-[350px] max-h-[400px] my-4"><canvas
id="yieldChart"></canvas></div></div><!-- Live Derivative Diagnostics Panel --><div id="diagnosticPanel"
class="bg-gradient-to-br from-slate-800 to-slate-900 text-white rounded-xl p-5 shadow-md border border-
slate-700 transition-all duration-300"><div class="flex items-center gap-2 mb-3"><div class="w-2.5 h-2.5
rounded-full bg-emerald-400 animate-pulse"></div><h4 class="text-xs font-semibold tracking-wider text-slate-
400 uppercase">Calculus Analysis Panel</h4></div><!-- Default Prompt / Empty State --><div
id="panelPrompt" class="text-center py-6 text-slate-400 text-sm"><p class="font-medium text-slate-300">No
tree age selected.</p><p class="text-xs mt-1">Click a point on the grid line above to run the derivative
calculation instantly.</p></div><!-- Active Diagnostic Content (Hidden initially) --><div id="panelContent"
class="hidden grid grid-cols-1 md:grid-cols-3 gap-4 items-center"><div class="bg-slate-700/40 p-3 rounded-lg
border border-slate-600/30"><span class="block text-[10px] text-slate-400 uppercase tracking-wide">Selected
Age ($x$)</span><div class="text-xl font-bold text-white font-mono mt-0.5"><span id="outAge">0</span>
<span class="text-xs font-normal text-slate-400">years old</span></div></div><div class="bg-slate-700/40 p-3
rounded-lg border border-slate-600/30"><span class="block text-[10px] text-slate-400 uppercase tracking-
wide">Total FFB Yield ($f(x)$)</span><div class="text-xl font-bold text-amber-400 font-mono mt-0.5"><span
id="outYield">0.00</span> <span class="text-xs font-normal text-slate-400">t/ha/yr</span></div>
</div><div class="bg-slate-700/40 p-3 rounded-lg border border-slate-600/30"><span class="block text-[10px]
text-slate-400 uppercase tracking-wide">Rate of Change ($f'(x)$)</span><div class="text-xl font-bold text-
emerald-400 font-mono mt-0.5"><span id="outDerivative">0.00</span> <span class="text-xs font-normal text-
slate-400">t/ha/yr</span></div></div></div><!-- Layman Explainer Box spans full width below key metrics --><div
class="md:col-span-3 mt-2 bg-emerald-950/40 p-4 rounded-xl border border-emerald-800/50"><span
class="text-xs font-bold text-emerald-400 block mb-1">🌴 Layman Biological Meaning:</span><p
id="outExplainer" class="text-sm text-slate-200 leading-relaxed"></p></div></div></div></div></div></main>
<!-- Footer Note --><footer class="bg-slate-100 border-t border-slate-200 py-4 text-center text-xs text-slate-
500 mt-8">Built for educational instruction inside Malaysian AgriTech and Food Security academic streams.
Uses dynamic Chart.js canvas render engines.</footer><!-- JavaScript Application Logic --><script>
// Setup structural data coordinates corresponding to plantation life cycles (Ages 1 to 28)
const yearsData = Array.from({length: 28}, (_, i) => i + 1);
// Core Cubic Function: f(x) = -0.015x³ + 0.52x² - 1.8x + 3.5
function calculateYield(x) {
  return (-0.015 * Math.pow(x, 3)) + (0.52 * Math.pow(x, 2)) - (1.8 * x) + 3.5;
}

// First Derivative Function: f'(x) = -0.045x² + 1.04x - 1.8
function calculateDerivative(x) {
  return (-0.045 * Math.pow(x, 2)) + (1.04 * x) - 1.8;
}

// Generate arrays matching the domain parameters
const yieldValues = yearsData.map(x => calculateYield(x));
// Locate UI DOM reference elements
const ctx = document.getElementById('yieldChart').getContext('2d');
const panelPrompt = document.getElementById('panelPrompt');
const panelContent = document.getElementById('panelContent');
const outAge = document.getElementById('outAge');
const outYield = document.getElementById('outYield');
const outDerivative = document.getElementById('outDerivative');
const

```

```

outExplainer = document.getElementById('outExplainer');// Compile and configure the ChartJS dual dataset
instancelet yieldChart = new Chart(ctx, {type: 'line',data: {labels: yearsData,datasets: [{label: 'Fresh Fruit Bunch
Yield f(x) (t/ha/yr)',data: yieldValues,borderColor: '#059669', // Emerald 600backgroundColor: 'rgba(5, 150,
105, 0.05)',borderWidth: 3,tension: 0.3,fill: true,yAxisID: 'y'}, {label: "Growth Velocity f'(x) (Rate of Change)",
data: derivativeValues,borderColor: '#38bdf8', // Sky 400borderWidth: 2,borderDash: [5, 5],tension: 0.3,fill:
false,yAxisID: 'yDerivative'hidden: false // Visible by default for analytical compariso}}],options: responsive:
true,maintainAspectRatio: false,plugins: {legend: position: 'top',labels: { font: { size: 12, weight: '500' },
usePointStyle: true }},tooltip: {enabled: true,mode: 'index',intersect: false}},// Trigger analytical recalculations
when user clicks on any grid pointonClick: (event, activeElements) => {if (activeElements.length > 0) {const
index = activeElements[0].index;const selectedAge =
yearsData[index];updateDiagnosticPanel(selectedAge);}},scales: {x: title: { display: true, text: 'Plantation Age
(Years)', font: { weight: 'bold' } },grid: { color: 'rgba(0,0,0,0.03)' },y: {type: 'linear',display: true,position:
'left',title: { display: true, text: 'Yield Output (Tonnes/Hectare/Year)', font: { weight: 'bold' } },min: 0max:
25,grid: { color: 'rgba(0,0,0,0.05)' },yDerivative: type: 'linear',display: true,position: 'right',title: { display: true,
text: 'Acceleration Velocity Vector', font: { weight: 'bold' } },grid: { drawOnChartArea: false }, // Avoid gridline
clutteringmin: -4,max: 6},// Fluid canvas initialization rendering animationsanimation: duration: 1600,easing:
'easeOutQuart'}}});// Business rules engine maps mathematics back into food security domain semantics
function updateDiagnosticPanel(age) {const yieldVal = calculateYield(age);const derivVal =
calculateDerivative(age);// Toggle visibility classes within layout container element
panelPrompt.classList.add('hidden');panelContent.classList.remove('hidden');// Hydrate metrics data readouts
outAge.innerText = age;outYield.innerText = yieldVal.toFixed(2);outDerivative.innerText = (derivVal >= 0 ? "+" :
"") + derivVal.toFixed(2);// Construct biological feedback copy based on the polarity and size of the derivative
vector let meaningString = "";if (age <= 3) {meaningString = `The derivative is **${derivVal.toFixed(2)}**. This
negative value reflects the juvenile establishment phase. Young trees are building root architecture and
metabolic resource sinks rather than producing collectable food output.`;} else if (age > 3 && age <= 7)
{meaningString = `The derivative is highly positive at **${derivVal.toFixed(2)}** t/ha/yr². This signifies the
**exponential expansion phase**. Yield output is skyrocketing annually. For food security pipelines, this
indicates high structural momentum where the trees are translating fertilizer blocks efficiently into canopy and
fruit mass.`;} else if (age > 7 && age < 11) {meaningString = `The derivative is stabilizing at
**${derivVal.toFixed(2)}** t/ha/yr². The rate of change is still positive, meaning yields are growing, but the
velocity is slowing down. Your plantation is approaching its peak carrying capacity.`;} else if (age >= 11 && age
<= 13) {meaningString = `The derivative is practically zero (**${derivVal.toFixed(2)}**). In calculus, $f'(x) = 0$
marks a critical local maximum. Biologically, this means your oil palms have reached **Prime Peak Maturity**.
Harvest output is maximized at roughly **${yieldVal.toFixed(1)} tonnes** per hectare. Management focus
here should shift completely from expansion stimulation to defensive preservation.`;} else if (age > 13 && age
<= 22) {meaningString = `The derivative has crossed the zero threshold into negative territory
(**${derivVal.toFixed(2)}**). Production efficiency is experiencing a slow decay. The palm structures are
growing excessively tall, moving active biological energy budgets toward baseline respiratory maintenance
rather than heavy bunch development.`;} else {meaningString = `The derivative drops to
**${derivVal.toFixed(2)}**. This deep negative slope indicates an **Agro-economic Replanting Alert**. Fruit
production velocity is declining steeply, and high tree heights make hand-harvesting inefficient. From a
national food security perspective, this block requires strategic systematic clearing and replanting with higher-
yielding clonal seeds.`;} outExplainer.innerHTML = meaningString;}</script></body></html>

```